Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-6

Number of questions: 15 **Total marks: 15 X 1= 15**

1. Which of the following temperature sensors generates an analog voltage proportional to the temperature?

   a. DS1621
   b. DHT11
   c. DS18B20
   d. LM 35

Answer: (d)

Justification: Please refer to Slides 15 of Lecture 1 of Week 6

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2. What are the key components of an IoT Sensing Device?

   a. Sensor, Processor, Wireless Communication Module, Power Management Unit
   b. Battery, Processor, Universal Serial Bus, Visual Display Unit
   c. Sensor, Compiler, Interpreter, Energy Harvester
   d. None of the above

Answer: (a)

Justification: Please refer to Slide 14 of Lecture 1 of Week 6

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3. What kind of sensor is ADXL335?

   a. Accelerometer
   b. Gyroscope
   c. Magnetometer
   d. Proximity sensor

Answer: (a)

Justification: Please refer to Slide 19 of Lecture 1 of Week 6

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4. “MOS gas sensor depend on the thermal energy for its operation which is supplied with a heater.” State True or False.

   a. True
   b. False

Answer: (a)

Justification: Please refer to Slide 4 of Lecture 2 of Week 6

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5. Which of the following is not a probable application of Gas Sensors?

a. Air quality monitoring
b. Leakage Detection of Toxic gases
c. Alcohol Breath Test
d. Motion Tracking

Answer: (d)

Justification: Please refer to Slide 8 of Lecture 2 of Week 6

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6. What is meant by the Sensitivity of a Gas Sensor?

a. Change in the output signal for a unit change in gas concentration
b. Ability to detect a particular gas in a mixture of different gases
c. Ability to provide consistent output over a long period of time
d. The time taken by the sensor to stabilize its response

Answer: (a)

Justification: Please refer to Slide 6 of Lecture 2 of Week 6

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7. The ModBus-TCP protocol defines two units in the data frame. One of them is Protocol Data Unit. What is the other one?

a. Analogous Data Unit
b. Advanced Data Unit
c. Application Data Unit
d. Architectural Data Unit

Answer: (c)

Justification: Please refer to Slide 6 of Lecture 3 of Week 6

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8. With respect to sensors, a camera which can simultaneously sense the ambient temperature as well as capture live video recordings is a type of _____ sensor.

a. Scalar
b. Multimedia

c. Hybrid
d. All of the above

Answer: (c)

Justification: Please refer to Page 105 of the book Introduction to IoT by S Misra, A Mukherjee, A Roy by Cambridge University Press

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9. EtherNet/IP defines two primary types of communications. What are those?

a. Explicit and Implicit
b. Internal and External
c. Inbound and Outbound
d. Telemetric and Geometric

Answer: (a)

Justification: Please refer to Slide 14 of Lecture 3 of Week 6

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10. Which of the following statements is incorrect about Profinet?

a. Profinet is the standard for industrial Ethernet
b. Profinet is based upon Ethernet/IP
c. Profinet can not be used for process control or process measurement
d. Profinet uses NRT, RT and IRT communication channels

Answer: (c)

Justification: Please refer to Slide 17 and 18 of Lecture 3 of Week 6

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11. Modbus-RTU is a serial protocol that follows the _____ architecture.

a. Producer/Consumer
b. Master/Slave
c. Scalar/Vector
d. None of the above

Answer: (b)

Justification: Please refer to Slides 15 and 16 of Lecture 3 of Week 6

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12. Which of the following network topologies is used by Profibus?

a. Ring Topology
b. Bus Topology
c. Mesh Topology

d. Star Topology

Answer: (b)

Justification: Please refer to Slide 5 of Lecture 4 of Week 6

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13. In the context of DeviceNet, CAN is the acronym for _____.

a. Cognitive Adaptable Network
b. Campus Area Network
c. Controller Area Network
d. Captive Area Network

Answer: (c)

Justification: Please refer to Slide 14 of Lecture 4 of Week 6

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14. How shorter TTL is achieved in uRLLC?

a. By using smaller slot length
b. By reducing Orthogonal Frequency Division Multiplexing symbols in TTL
c. Reducing symbol duration
d. All of the above

Answer: (d)

Justification: Please refer to Slide 23 of Lecture 5 of Week 6

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15. What is the frequency spectrum of mmWave Communication?

a. 30 – 300 MHz
b. 550 – 1720 KHz
c. 88 – 108 MHz
d. 30 – 300 GHz

Answer: (d)

Justification: Please refer to Slide 26 of Lecture 5 of Week 6
